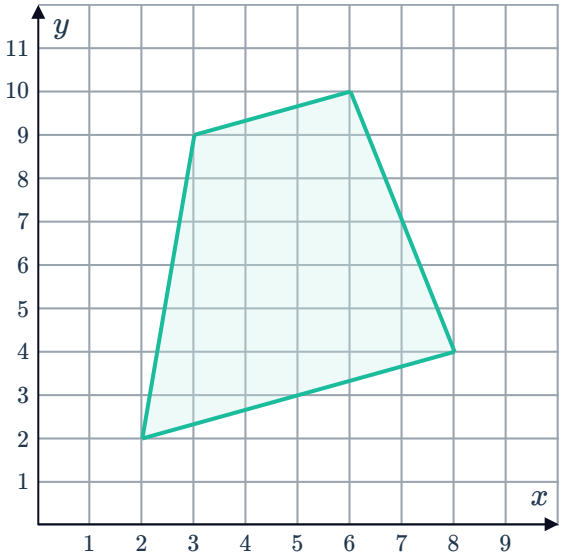
Optimising using corner points principle

1 The following graph shows a region of feasible solutions for the objective function $z = 4x + 3y$:



a Calculate the value of the function for each of the following corner points:

i $(2, 2)$ **ii** $(3, 9)$ **iii** $(6, 10)$ **iv** $(8, 4)$

b Calculate the maximum value of the objective function and state at what point it is obtained.

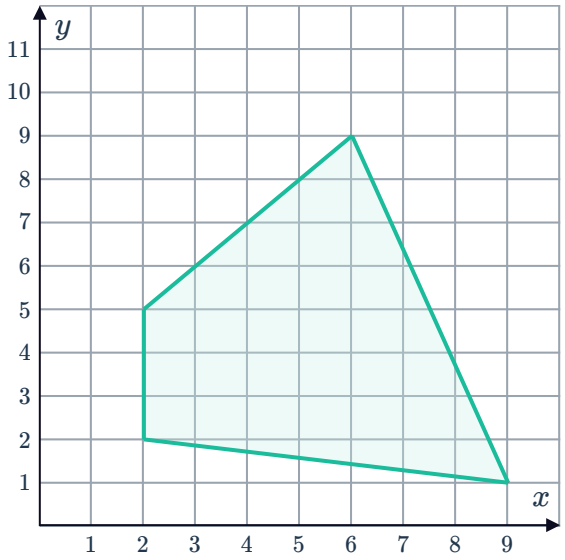
c Calculate the minimum value of the objective function and state at what point it is obtained.

2 For each of the following graphs of feasible regions and corresponding objective functions:

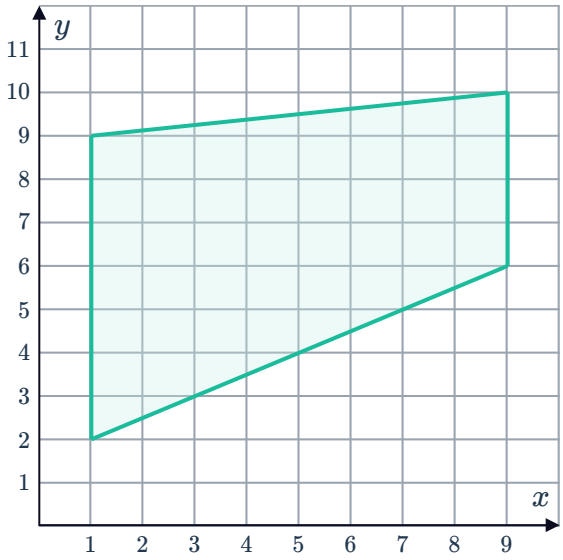
i Calculate the maximum value of the objective function and state at what point it is obtained.

ii Calculate the minimum value of the objective function and state at what point it is obtained.

a The objective function is $z = 5x + 2y$.

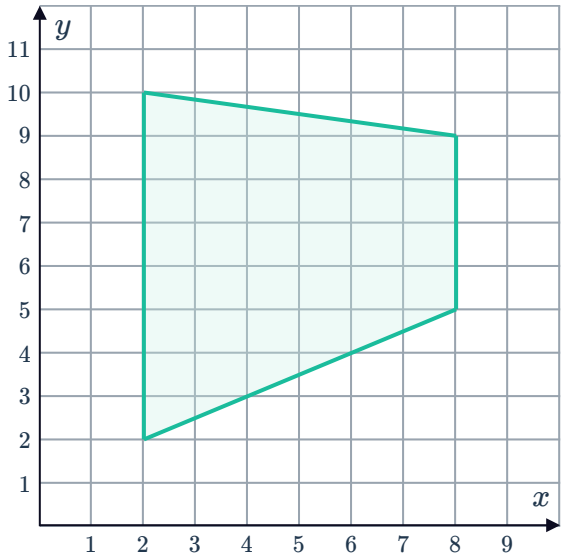


b The objective function is $z = 3x + 6y$.

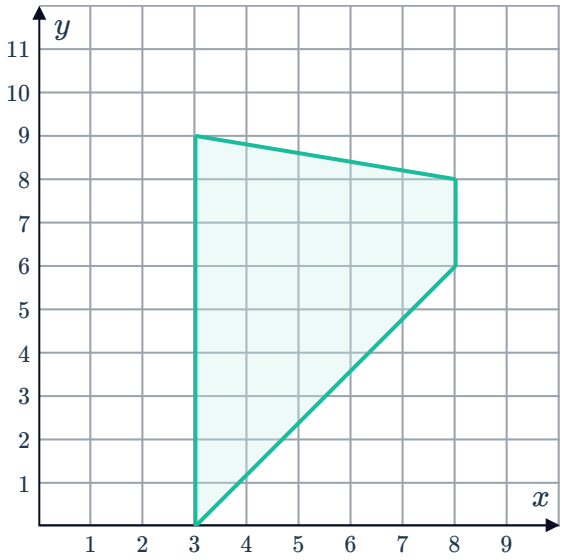


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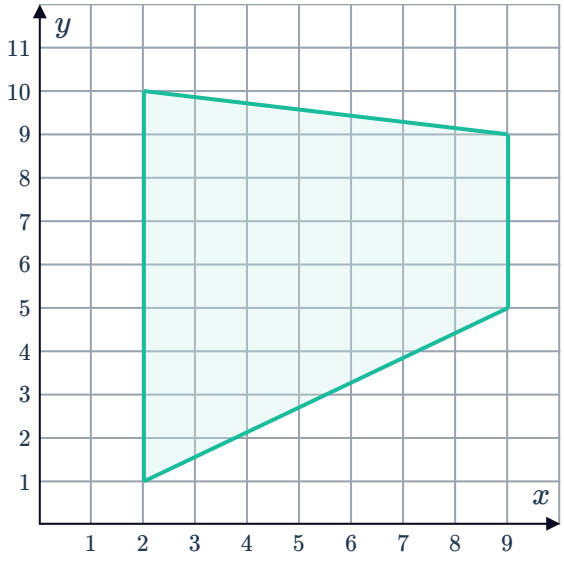
c The objective function is $z = 3x + 3y$.



d The objective function is $10y$.



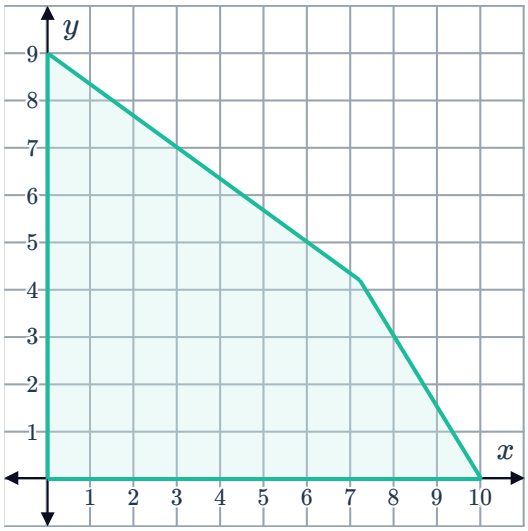
e The objective function is $z = 5x - 2y$.



3 The following graph represents the feasible region for the objective function

$$T = 14x + 31y$$

Given that the coordinates of the corner vertices are $(0, 9)$, $\left(\frac{36}{5}, \frac{21}{5}\right)$ and $(10, 0)$, determine the maximum value of the objective function T and state at what point it is obtained.



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4 For each of the following systems of inequalities:

i Plot the feasible region defined by the four inequalities.

ii Write the coordinates of the vertices of the region defined by the four inequalities.

a $x \geq 0$, $y \geq 0$, $x \leq 4$, and $y \leq 9$.

b $x \geq 0$, $y \geq 0$, $x < 4$, and $y \leq 7 + x$.

c $x \geq 0$, $y \geq 0$, $x + y < 7$, and $12x - 12y < 24$.

d $x > 3$, $y \geq 0$, $4x + y \leq 20$, and $y \leq 11 - x$

5 For each of the following objective functions and their constraints:

i Graph the region defined by the constraints in the first quadrant.

ii Calculate the maximum value of the function P and state at what point it is obtained.

iii Calculate the minimum value of the function P and state at what point it is obtained.

a Objective function $P = -28x + 10y + 56$, subject to the constraints $7x + 5y \leq 35$, $0 \leq x \leq 2$ and $0 \leq y \leq 6$.

b Objective function $P = 12x + 46y$, subject to the constraints $4x + 3y \leq 24$, $3x + 4y \leq 28$, $x \geq 0$ and $y \geq 0$.

6 For each of the following objective functions and their constraints:

i Graph the region defined by the constraints in the first quadrant.

ii Determine the maximum value of the objective function and the values of x and y for which the maximum occurs.

iii Check your answer using the sliding line method.

a Objective function $z = 6x + 2y$, subject to the constraints $6x + 2y \leq 60$, $2x + 6y \leq 54$, $x \geq 0$ and $y \geq 0$.

b Objective function $z = 22x + 7y$, subject to the constraints $x + 2y \geq 8, x + y \geq 6$, $x + y \leq 11, x \geq 0$, and $y \geq 0$.

c Objective function $z = 9x + 14y$, subject to the constraints $2x + y \geq 8, x + 2y \geq 8$, $x + y \leq 8, x \geq 0$ and $y \geq 0$.

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Applications

7 A manufacturer produces two types of tables. Each table requires a cabinet maker and a painter to build. The time taken for each worker varies according to the table below:

	Cabinet Maker	Painter
Round Table	2	2
Square Table	3	2
Total time available in a week	36	32

Let x represent the number of round tables built in a week, and let y represent the number of square tables built in a week.

a Construct the set of four constraint inequalities.

b Graph the region defined by the contraints in the first quadrant.

c Hence, list the four vertices of the feasible region.

d If a round table sells for a profit of $\$240$ and a square table makes a profit of $\$280$, write the objective function that models the weekly profit P.

e Using the sliding line or corner point method, determine the vertex that maximises the manufacturer's profits.

f Calculate the maximum weekly profit.

8 A shoe manufacturer produces shoes for both men and women. Each pair of shoes is made of rubber for the soles and leather for the upper. The amount of material required, in decimetres cubed, to make each type is shown in the table below:

	Rubber (dm^3)	Leather (dm^3)
Women's Shoe	8	8
Men's Shoe	16	12
Maximum amount of available materials in a week	3520	3360

Let x represent the number of pairs of women's shoes made in a week, and let y represent the number of pairs of men's shoes made in a week.

a Construct the set of four constraint inequalities.

b Graph the region defined by the constraints in the first quadrant.

c List the four vertices of the feasible region.

d If one pair of women's shoes sells for a profit of $\$15$ and one pair of men's shoes makes a profit of $\$26$, write the objective function that models the weekly profit P.

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e Using the sliding line or corner point method, determine the vertex that maximises the manufacturer's profit.

f Hence state the maximum weekly profit.

9 The popular shop Bergner's Burgers makes and sells two types of burgers; beef and chicken. Each day, Mr Bergner knows that they need to make at least 190 beef burgers and 170 chicken burgers. The maximum number of burgers that can be made in one day is 460.

Let x represent the number of beef burgers made per day, and let y represent the number of chicken burgers made in a day.

a Construct the set of three constraint inequalities.

b Graph the region defined by the inequalities.

c If Bergner's Burgers makes a profit of $\$1.70$ from each beef burger, and a profit of $\$1.00$ from each chicken burger, write the objective function that models the weekly profit P. Assume that every burger made is sold.

d Evaluate the profit corresponding to each vertex of the feasible region.

e State the maximum daily profit.

f State the number of beef burgers and number of chicken burgers that should be made each day to achieve the maximum profit.

g If the prices of the burgers were switched, such that the profit from beef burgers was $\$1.00$ and the profit from chicken burgers was $\$1.70$, how many of each type of burger should now be made for a maximum profit?

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